Time Series Analysis and Forecasting of Weekly RSV Emergency Department Visits in the United States

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STA 137 Project

Abstract

Respiratory Syncytial Virus (RSV) is a common respiratory infection that produces seasonal outbreaks and contributes to increased emergency department utilization during peak periods. Understanding the temporal behavior of RSV activity is important for public health monitoring and healthcare preparedness. In this study, weekly RSV emergency department visits percentages in the United States are analyzed using classical time series methods. The data are first explored through visualization and summary statistics to identify key patterns such as seasonality and variability. A classical additive decomposition is then used to separate the series into trend, seasonality, and residual components. After removing the smooth structure of the series, the remaining stationary components are modeled using autoregressive moving average (ARMA) models. Forecasts of future RSV emergency department visit percentages are obtained by combining predictions of the smooth and rough components. The analysis highlights the strong seasonal nature of RSV outbreaks and provides short-term forecasts of expected RSV activity.

Introduction

Respiratory Syncytial Virus (RSV) is a major cause of respiratory illness and often leads to seasonal increases in healthcare utilization. RSV outbreaks typically occur during winter months and can place significant strain on healthcare systems, particularly emergency departments. Monitoring the temporal behavior of RSV activity is therefore important for public health planning and hospital preparedness.

Time series analysis provides a statistical framework for studying patterns that change over time and for producing forecasts of future observations. By decomposing a time series into interpretable components such as trend, seasonality, and random variation, it is possible to understand the underlying structure of the data and model remaining dependence.

The goal of this study is to analyze weekly RSV emergency department visit percentages in the United States and to develop a statistical model able to forecast future RSV activity. The analysis focuses on identifying seasonal outbreak patterns, examining the trend of RSV activity over time, and modeling the remaining dependence using ARMA methods. Accurate short term forecasts of RSV activity can help hospitals anticipate increases in emergency department demand and assist public health agencies in preparing for seasonal outbreaks.

The remainder of the paper is organized as follows. First, the dataset and its main characteristics are described. The time series is analyzed through visualization, transformation checks, and classical decomposition to identify smooth components of the series. Finally, the

residual component is modeled using an ARMA process and used to generate forecasts of future RSV emergency department visit percentages.

Data Description

The data used in this analysis comes from the Centers for Disease Control and Prevention (CDC) National Syndromic Surveillance Program (NSSP). The dataset is part of the CDC Respiratory Virus Response surveillance system and contains weekly percentages of emergency department visits associated with respiratory viruses, including COVID-19, influenza, and Respiratory Syncytial Virus (RSV). The data are publicly available through the CDC public health surveillance portal.

Each observation represents the percentage of emergency department visits attributed to a specific respiratory pathogen during a given week and geographic region. The dataset contains four variables: the week ending date, the pathogen type, the geographic location, and the percentage of emergency department visits associated with that pathogen

For this analysis, the data were filtered to include only observations where the pathogen was RSV. This filtering produces a weekly time series representing the national proportion of emergency department visits associated with RSV.

The weekly RSV visit percentages range from values close to zero during periods of minimal virus circulation to peaks exceeding three percent during seasonal outbreaks. Figure 1 displays the time series plot of the weekly RSV emergency department visit percentages. The series reveals clear seasonal behavior, with peaks occurring during winter months and much lower levels during the summer.

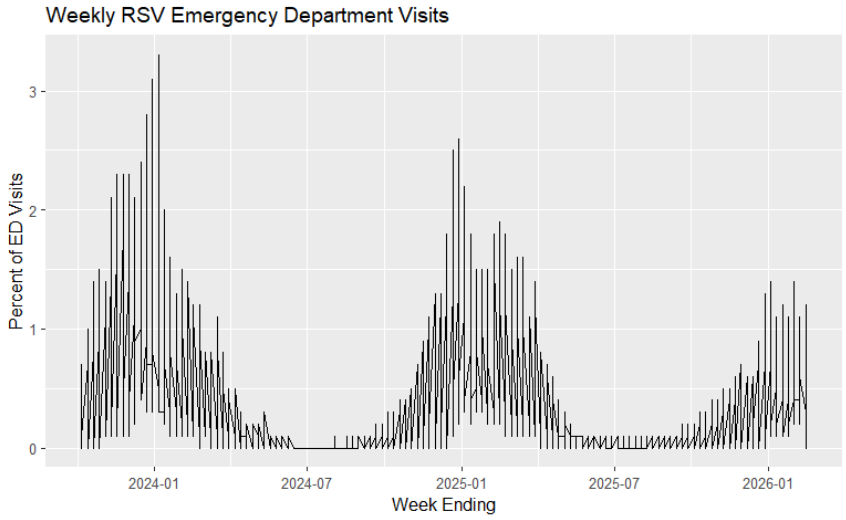


Figure 1. Time Series

The distribution of RSV percentages is shown in the histogram in Figure 2. Most observations occur at relatively low values, reflecting weeks with low RSV activity, while a smaller number of observations correspond to outbreak peaks. This produces a right-skewed distribution.

These characteristics indicate that the RSV time series contains strong seasonal patterns along with short-term fluctuations, making it well suited for classical time series analysis and decomposition methods.

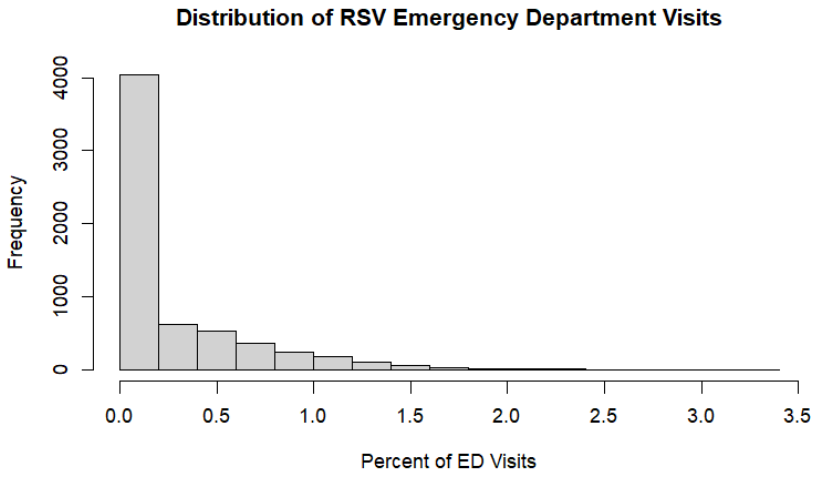


Figure 2. Histogram

Data Analysis

Initial Data Transformations

The time series was first examined to determine whether a transformation was necessary to stabilize the variance or improve symmetry before applying classical decomposition methods. The time series plot weekly RSV emergency department visit percentages reveal tall seasonal spikes, with peaks occurring roughly once per year during winter months. Between outbreak periods the series remains close to zero, indicating prolonged periods of low RSV activity.

The histogram of the series shows that the distribution of weekly RSV visit percentages is right skewed. Most observations occur at relatively low values, while a smaller number of observations correspond to large outbreak peaks. This skewness reflects the epidemiological behavior of RSV, where outbreaks produce sharp increases in case activity during limited periods of the year.

To further assess whether a transformation might be beneficial, the Box–Cox transformation parameter was estimated. The estimated value was $\lambda = 0.406$, which suggests that a mild power transformation could potentially stabilize the variance. However, inspection of the time series plot indicates that the variability of the series remains reasonably consistent across time, and the observed skewness primarily results from occasional outbreak spikes rather than persistent variance instability. For this reason, the subsequent decomposition analysis was performed on the original data scale.

Analysis of the Smooth Component

To identify the underlying structure of the series, the time series was decomposed into observed, trend, seasonal, and random components using classical additive decomposition shown in Figure 3. The observed series is represented as the sum of a smooth component consisting of trend and seasonality and remaining stochastic component.

The observed component represents the original RSV emergency department visit percentages over time. It combines the effects of the long term trend, the repeating seasonal cycle, and short-term random fluctuations.

The seasonal component reveals a strong and regular yearly cycle in RSV activity. The estimated seasonal pattern shows that RSV emergency department visit percentages consistently increase during winter months and decline during the summer. This recurring pattern aligns with known seasonal transmission patterns of RSV in temperate climates.

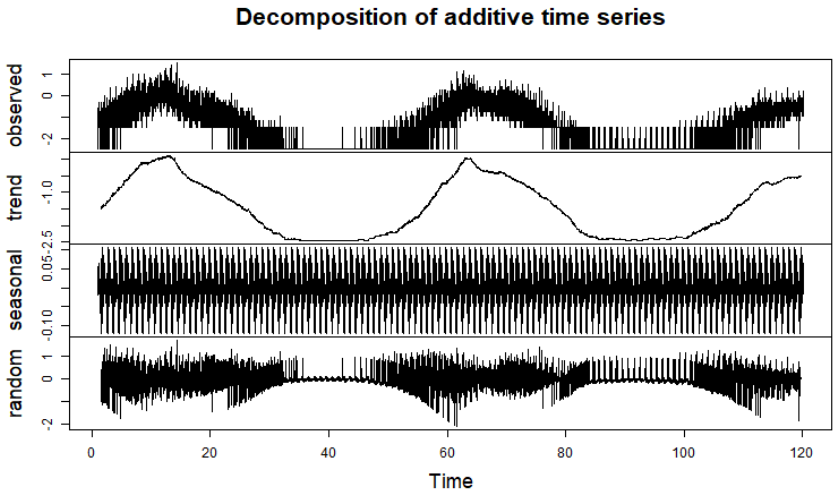


Figure 3. Decomposition

The trend component captures slower changes in the overall level of RSV activity across the observed time period. While the magnitude of outbreaks varies slightly from year to year, the dominant structure of the series remains the seasonal cycle rather than a persistent long term upward or downward trend.

The random component represents the remaining variation after removing the trend and seasonal structure. These fluctuations correspond to short-term deviations from the expected seasonal pattern and capture irregular variation that cannot be explained by the smooth component alone.

To assess how well the smooth component explains the observed data, the observed series was plotted together with the estimated smooth component shown in Figure 4. It is defined as the sum of the trend and seasonal components. The smooth component closely follows the general

shape of the outbreak cycles, capturing both the timing and magnitude of the major seasonal peaks.

After removing the smooth component, the remaining residual series represents the rough component of the time series. These residual fluctuations reflect short-term deviations from the underlying seasonal pattern and are assumed to be approximately stationary. This residual series will be examined further and modeled using ARMA processes in order to capture any remaining temporal dependence.

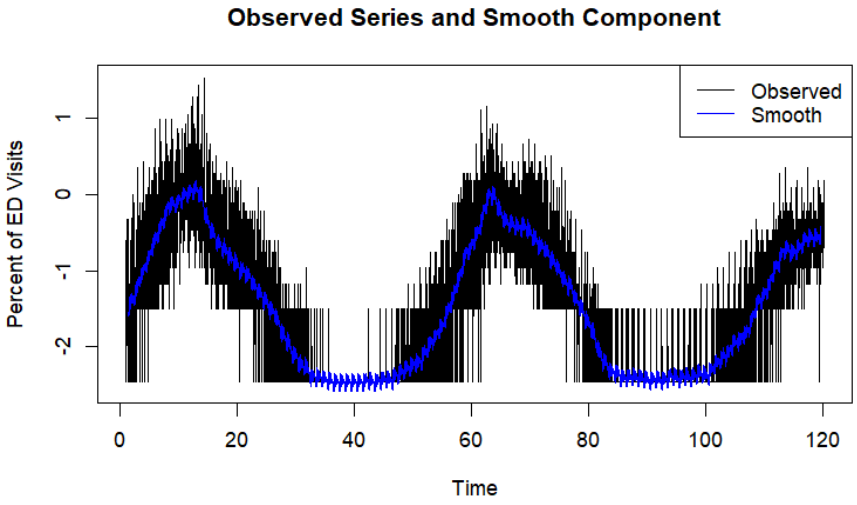


Figure 4. Smooth Component

Analysis of the Residual

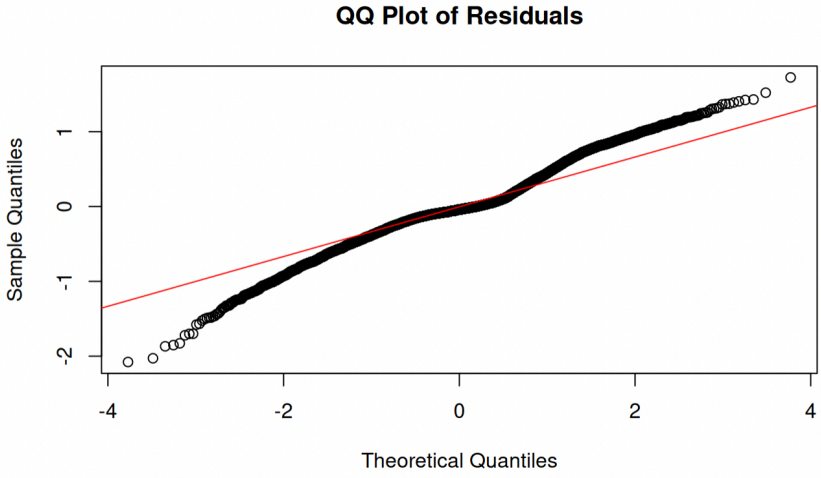


Figure 5. QQ Residual

After removing the smooth component, the remaining residual series was examined to determine whether it behaves like a stationary random process and whether additional time

dependence remains. Diagnostic plots were used to evaluate the distributional properties and dependence structure of the residuals.

Figure 5 displays the QQ plot of residuals. In this case, most of the points follow the line reasonably well in the center of the distribution, while small deviations appear in the tails. These departures suggest that the residuals are approximately normal but may contain slightly heavier tails than a perfect normal distribution. This behavior is common in epidemiological time series where occasional spike outbreaks can create extreme observations.

This result indicates that the normality assumption is reasonable, although some minor deviations remain. Because ARMA modeling primarily requires stationarity and dependence structure rather than perfect normality, the residual series remain suitable for further time series modeling.

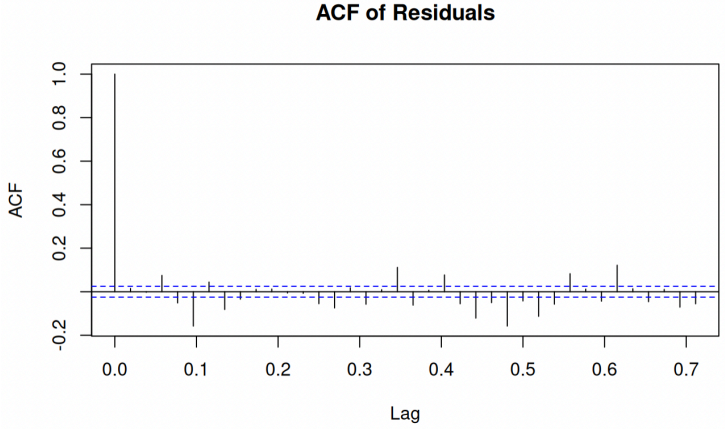


Figure 6. ACF Residual

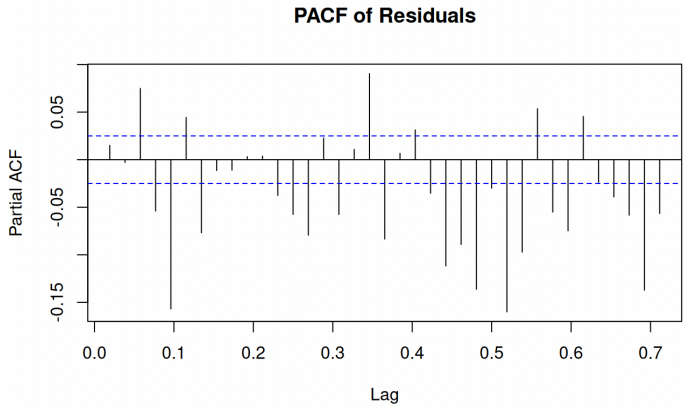


Figure 7. PACF Residual

Figure 6 and Figure 7 display the autocorrelation function (ACF) and partial autocorrelation function (PACF) of the residual series obtained after removing the trend and seasonal components. These plots are used to examine whether the residuals behave like white noise or whether additional temporal dependence remains in the series.

The dashed horizontal lines in the ACF plot represent approximately 95 percent confidence bounds given by ±1.96 divided by the square root of n. If the residuals were independent, about 95 percent of the sample autocorrelations would be expected to lie within these bounds. While many of the autocorrelations are small, several spikes extend beyond the confidence limits, indicating that some dependence remains in the residual series.

The PACF plot shows a similar pattern, with several partial autocorrelations exceeding the significance bounds. These spikes suggest that the residuals still contain short-term lagged dependence that has not been fully removed by the trend and seasonal decomposition.

Together, the ACF and PACF diagnostics indicate that the residuals do not completely conform to white noise. This remaining dependence motivates the use of ARMA models to capture the stochastic structure of the rough component of the RSV time series.

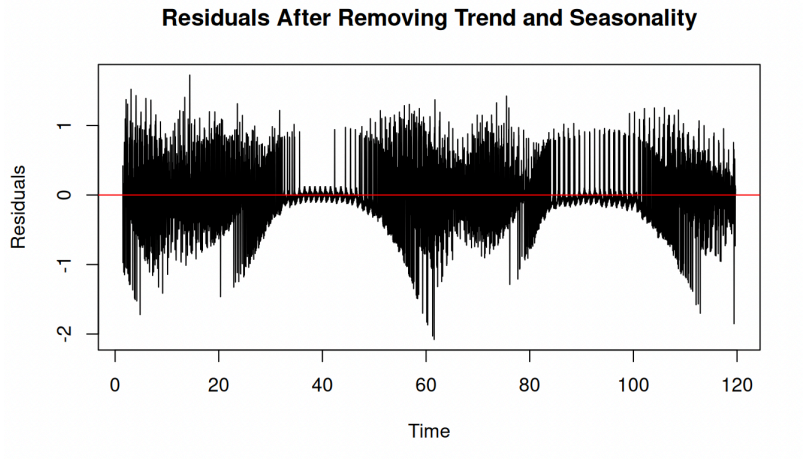
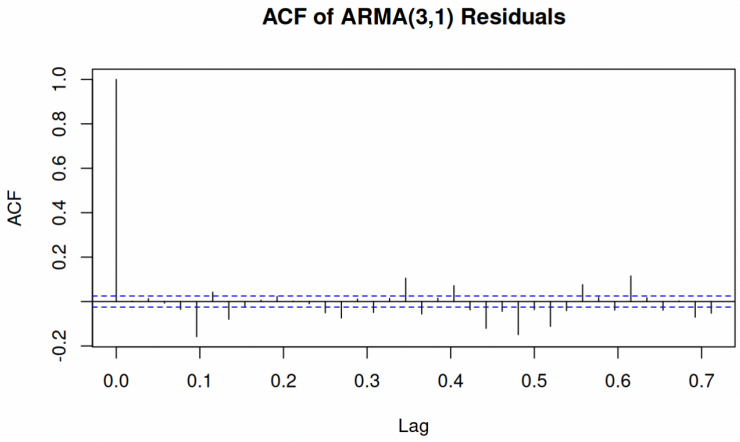


Figure 8. Residual Plot

Figure 8 shows the residual series after removing the estimated trend and seasonal components. The residuals fluctuate around zero with no clear long term pattern, which supports the assumption that the smooth structure of the series has been successfully removed. The variation appears relatively stable across time, although occasional larger deviations occur during periods corresponding to outbreak transitions. These fluctuations represent short-term irregular behavior that is not explained by the deterministic trend and seasonal components. Overall, the residual series appears approximately stationary and centered around zero, which is consistent with the assumptions required for ARMA modeling. Combined with the ACF and PACF diagnostics, this residual plot indicates that the remaining variation behaves like a stationary stochastic process. Therefore, modeling the residual component using an ARMA process is appropriate for capturing the remaining temporal dependence in the RSV time series.

Analyzing the rough component



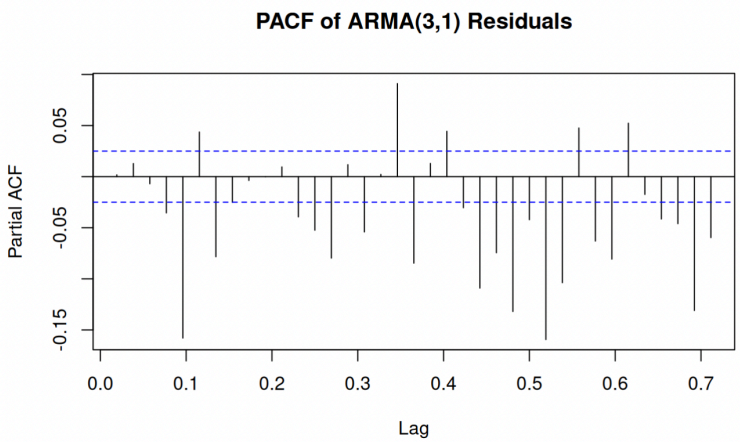


Figure 9. ACF ARMA Residual

Figure 10. PACF ARMA Residual

Several ARMA models were fitted to the residual series in order to capture the remaining dependence in the rough component. The candidate models included AR(1), MA(1), ARMA(1,1), ARMA(2,1), ARMA(2,2), and ARMA(3,1). These models were compared using the Akaike Information Criterion (AIC). Among the candidate models, the ARMA(3,1) model was selected because it produced the lowest AIC value, indicating the best balance between model fit and model complexity.

After fitting the ARMA(3,1) model, diagnostic checks were performed to evaluate the behavior of the model residuals. Figure 9 and Figure 10 show the ACF and PACF plots of the ARMA(3,1) residuals. Most of the autocorrelations fall within the confidence bounds, and no strong pattern of dependence is visible across lags. This suggests that the ARMA model has captured most of the remaining temporal structure in the residual series.

A Ljung–Box test was also applied to further assess whether the residuals behave like white noise. The test produced $X^2 = 439.52,\ df = 20,\ p < 0.001$, indicating that the null hypothesis of white noise is rejected and that some autocorrelation may still remain in the residual series.

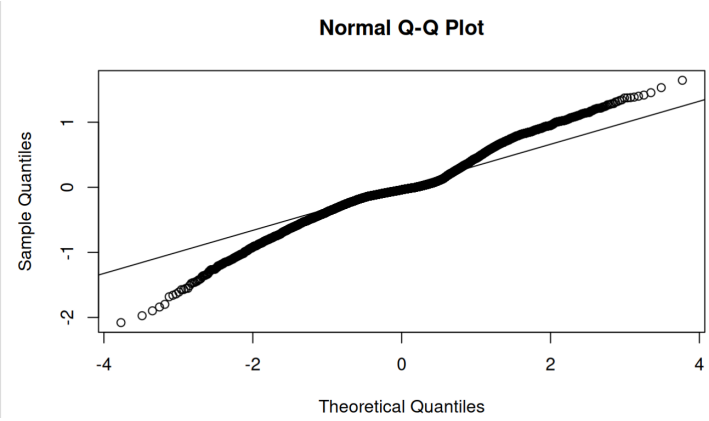


Figure 11. QQ ARMA Residual

Finally, a normal Q-Q plot was used to assess whether the residuals are approximately normally distributed. Figure 11 shows the Q-Q plot of the ARMA(3,1) residuals. The points follow the reference line reasonably well for most of the distribution, although some deviations appear in the extreme tails. This indicates that the residuals are approximately normal with slight departures at larger values. Overall, these diagnostic results suggest that the ARMA(3,1) model provides a reasonable representation of the rough component of the RSV time series.

Predict future values

Future values of the RSV time series were predicted by combining forecasts of the smooth component and the rough component. The smooth component consists of the estimated trend and seasonal effects obtained from the decomposition of the series.

The trend component was forecast by fitting a linear regression model to the estimated trend values as a function of time and projecting the fitted trend forward for the next 20 weeks. The seasonal component was predicted by repeating the most recent seasonal cycle, since the data exhibit a yearly seasonal pattern with a period of 52 weeks. The predicted smooth component was then obtained by adding the forecasted trend and seasonal components.

The rough component was modeled using the ARMA(3,1) process selected in the previous section. Forecasts of the residual process were obtained using the ARMA prediction algorithm implemented in R, which follows the innovations based forecasting procedure described in Section 3.5 of the lecture notes.

The final forecast of weekly RSV emergency department visit percentages was obtained by adding the predicted smooth component and the predicted rough component. Figure 12 displays the resulting forecasts for the next 20 weeks together with the observed series and the associated 95 percent prediction intervals.

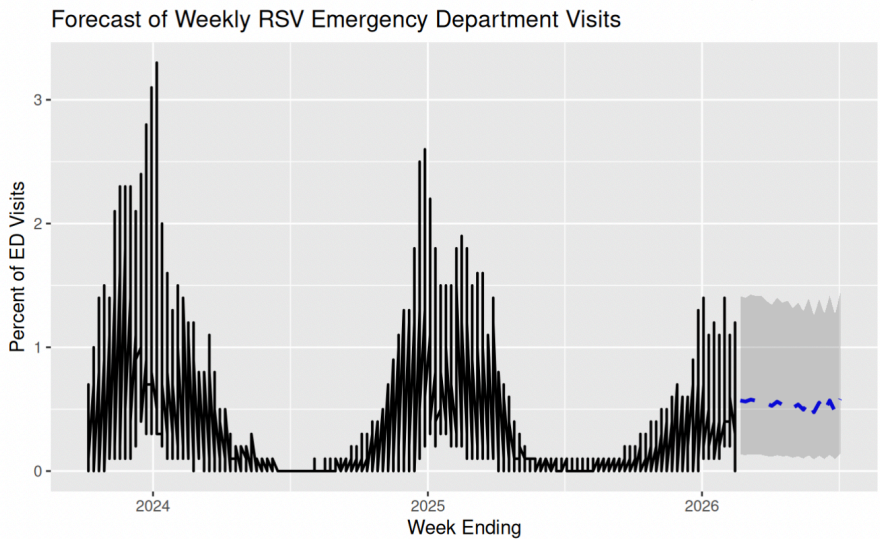


Figure 12. Forecast

Discussion

The analysis shows that weekly RSV emergency department visit percentages display a strong and consistent seasonal pattern, with peaks occurring during winter months and very low activity during the summer. The classical decomposition successfully separated the series into trend, seasonal, and residual components, and the seasonal cycle explains most of the variation observed in the data.

After removing the smooth component, the remaining residual series appeared approximately stationary. The ACF and PACF plots indicated that short term dependence remained in the data, which motivated the use of ARMA models to capture the stochastic structure of the rough component. Although the ARMA(3,1) model reduced much of the autocorrelation present in the residual series,the Ljung–Box test indicates that some autocorrelation may still remain. This suggests that the ARMA model may not fully capture all temporal dependence in the series.

One limitation of the analysis is that the Ljung–Box test indicates remaining autocorrelation in the residuals, suggesting that additional model structure could potentially improve the fit. In addition, the forecasting approach assumes that the seasonal pattern and model structure remain stable over time. Despite these limitations, the model captures the primary temporal patterns in the data and provides reasonable short-term forecasts of RSV activity.

Conclusion

This study analyzed weekly RSV emergency department visit percentages in the United States using classical time series methods. The goal of the analysis was to understand the temporal behavior of RSV activity and to develop a model capable of forecasting future observations. The time series exhibited a strong seasonal pattern, with outbreaks occurring during winter months and much lower activity during the summer.

Classical decomposition allowed the series to be separated into trend, seasonal, and residual components, revealing that the seasonal cycle explains most of the variation in RSV activity. After removing the smooth structure, the remaining residual variation was modeled using an ARMA(3,1) process. The model captures the short term dependence in the data and was used to produce forecasts of future RSV activity.

Overall, the results highlight the predictable seasonal nature of RSV outbreaks and demonstrate how time series methods can be used to analyze and forecast public health trends.

References

Centers for Disease Control and Prevention (CDC). National Syndromic Surveillance Program (NSSP). Respiratory Virus Response Surveillance Data. https://www.cdc.gov

Hyndman, R. J., & Athanasopoulos, G. (2021). Forecasting: Principles and Practice. OTexts.